

Quiz — 1.4.3 Boolean Algebra

Name: _____ Date: _____ **Mark:** _____ / 30

OCR H446 · Component 01 · 1.4.3 — show all working

Section A — Multiple choice (8 marks)

1. For which input combination does a 2-input **AND** gate output 1? A. A=0, B=0 B. A=1, B=0 C. A=0, B=1 D. A=1, B=1 Your answer: ☐
2. Which logic gate outputs 1 **only when its two inputs differ**? A. AND B. OR C. XOR D. NOR Your answer: ☐
3. Which two gates are described as **universal** (any circuit can be built from them alone)? A. AND and OR B. NAND and NOR C. XOR and NOT D. OR and NOT Your answer: ☐
4. De Morgan's law states that $\neg(A + B)$ is equivalent to: A. $\neg A + \neg B$ B. $\neg A \cdot \neg B$ C. $A \cdot B$ D. $\neg(A \cdot B)$ Your answer: ☐
5. Which Boolean identity is $\neg(\neg A) = A$? A. Idempotent B. Absorption C. Double negation D. Complement Your answer: ☐
6. In a **half adder**, the Sum output is given by: A. $A \cdot B$ B. $A + B$ C. $A \oplus B$ D. $\neg(A \cdot B)$ Your answer: ☐
7. What is the role of a **D-type flip-flop**? A. To add two bits together B. To store one bit of data on a clock edge C. To invert its input D. To convert binary to hexadecimal Your answer: ☐
8. When grouping 1s on a Karnaugh map, groups must be: A. Always exactly size 3 B. As small as possible C. Powers of two (1, 2, 4, 8...) and as large as possible D. Only ever a single cell Your answer: ☐

Section B — Calculations & short answer (17 marks)

9. Complete the truth table for the expression $Q = \neg(A \cdot B) + (A \cdot C)$. [5]

A	B	C	$A \cdot B$	$\neg(A \cdot B)$	$A \cdot C$	Q
0	0	0				
0	0	1				
0	1	0				
0	1	1				
1	0	0				
1	0	1				
1	1	0				
1	1	1				

10. Use **De Morgan's law** (and double negation) to simplify $\neg(\neg A + B)$ to an expression with no outer bar. Show each step. [3]

Working space:

11. A **half adder** receives inputs **A = 1** and **B = 1**. State the **Sum** and **Carry** outputs and the Boolean expression for each. [3]

Working space:

12. Complete this truth table for a **half adder** for all input combinations. [3]

A	B	Sum ($A \oplus B$)	Carry ($A \cdot B$)
0	0		
0	1		
1	0		
1	1		

13. Simplify the expression $A \cdot (A + B)$ using a Boolean identity, and name the law used. [3]

Working space:

Section C — Exam-style question (5 marks)

14. A **full adder** adds three input bits: A, B and a carry-in (C_{in}), producing a Sum and a carry-out (C_{out}).

(a) Give the Boolean expression for the **Sum** output of a full adder. [1]

(b) Give the Boolean expression for the **Cout** output of a full adder. [2]

(c) Explain how a full adder differs from a half adder, and why full adders are needed to add multi-bit binary numbers. [2]

Working space: